

CSE Project Proposal

Centralized Research Workspace (CRW)

Course: Web Architecture (CSE 4636) — **Team Name:** noWayHome
Department: Computer Science and Engineering (BSc in CSE)

1. Team Information

Team Members	Student ID
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2. Summary of the Idea

Modern research workflows suffer from fragmentation across disjointed platforms for communication, resource sharing, and writing. Centralized Research Workspace (CRW) is a dedicated, all-in-one web platform designed to eliminate "app-switching fatigue." By consolidating literature management, task tracking, manuscript development, and meeting coordination, CRW streamlines the academic research lifecycle from project inception to publication.

3. Key Features

- **User & Team Management:** Secure authentication, distinct team creation, and role-based access for concurrent projects.
- **Literature & Knowledge Base:** Centralized paper repository featuring collaborative annotations and shared background research logs.
- **Project & Task Tracking:** Built-in to-do lists, automated reminders, and visual milestone scheduling.
- **Manuscript Development:** A progressive, Markdown-integrated workspace for drafting and organizing academic papers.
- **Meeting Coordination:** Integrated logging for meeting minutes, key decisions, and team action items.

4. Unique Selling Point (USP)

Unlike generic productivity tools (e.g., Trello, Notion), CRW is strictly research-oriented. Every feature is purpose-built for the academic lifecycle, providing a tailored environment that aligns perfectly with how researchers actually read, write, and collaborate, rather than forcing them to adapt generic platforms.

5. Technologies

CRW will utilize a modern, scalable full-stack architecture centered around a **Spring Boot** backend:

- **Backend Core: Java & Spring Boot** for rapid development, robust RESTful APIs, and enterprise-level stability.
- **Database & ORM: PostgreSQL** (or MySQL) managed via **Spring Data JPA/Hibernate** for efficient relational mapping of workspaces and metadata.
- **Security: Spring Security & JWT** for stateless, role-based access control (e.g., Admin, Researcher).
- **Frontend SPA: React.js or Vue.js** to deliver a responsive, fluid user interface that minimizes page reloads and workflow friction.